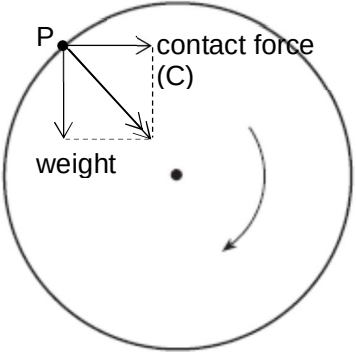


	Passes through (0.40, 0.80). Accuracy up to half small square	A1
2a	$\omega = 2\pi (N/t)$ $= 2\pi (100/60)$ $= 10.472 \text{ rad s}^{-1}$ $F = mr\omega^2$ $= (3.8 \times 10^{-3}) (0.118) (10.472)$ $= 4.9172$ $= 4.9 \times 10^{-2} \text{ N}$	M1 M1 A1
2b	 Arrow C must be equal length to weight and perpendicular to weight.	A1

3

2ci	Direction of weight and contact force is opposite to each other Contact force is the largest (at the lowest point) Centripetal force is upwards Therefore, plasticine is most likely to first lose contact with the disc at the lowest point of the revolution.	M1 M1 M1 A0
2cii	At lowest point, $C_{\max} - mg = mr\omega^2$ $0.23 - (3.8 \times 10^{-3})(9.81) = (3.8 \times 10^{-3})(0.118) \omega^2$ $\omega = 20.73 = 21 \text{ rad s}^{-1}$	C1 A1
2ci	Temperature at which all substances have a minimum (internal/kinetic) energy	B1